

VBF-T1LC-DVPR-01

Design Verification Plan & Report

Case: t1_lc | Date: 2026-08-26

Prepared: _____ Reviewed: _____ Approved: _____

Design Verification Plan & Report (virtual test) — VBF-T1LC-DVPR-01

One row per verification item; result values mechanically taken from simulation outputs.

Item	Condition	Result	Determination	Source
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1C discharge capacity	1C CC discharge to 2.5 V, 25 deg C	5.030 Ah	[PASS] (vs 5.0 Ah nominal)	cell/r3_V3_1c_dfn.json:capacity_ah
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Energy density	1C discharge energy / contract mass	413.02 Wh/kg	[PASS] (≥ 392.61 Wh/kg)	cell/r3_V3_energy.json:energy_density_wh_kg
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4C fast-charge temperature rise	4C charge @45 deg C, lumped thermal	328.56 K (55.4 deg C)	[PASS] (≤ 333.15 K)	cell/r3_V3_4c_dfn.json:T_max_K
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4C fast-charge plating	4C charge @45 deg C + plating module	anode_potential_v min +0.0368 V	[PASS] (no value < 0 V)	cell/r3_V3_4c_dfn.json:anode_potential_v
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Overcharge to 4.7 V thermal runaway	0.5C charge to 4.7 V -> run-tr	triggered = false	[PASS] (no TR)	cell/r3_V3_tr.json:triggered
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Voltage window	upper/lower cut-off	4.2 V / 2.5 V	[PASS] (per design)	parameter set
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Nail penetration	—	N/A (beyond pure simulation boundary, requires physical experiment)	N/A	—
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Crush / drop / external short	—	N/A (requires physical experiment)	N/A	—
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Cycle life (aging)	—	N/A (aging model not run for this case)	N/A	—
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Rate-pulse internal resistance	—	N/A (not in simulation protocol set)	N/A	—
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Conclusion

- Passed virtual verification: 1C capacity, energy density, 4C temperature rise, 4C plating, overcharge-to-4.7 V thermal runaway, voltage window.

- Uncovered (paper limitations): nail penetration, crush/drop, cycle-life aging, rate-pulse DCR — require physical experiment or aging simulation.